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1 # Установка необходимых библиотек
2 !pip install git+https://github.com/jaredwinick/img2vec-keras.git
3 !pip install gdown
4
5 import gdown
6 import glob
7 import numpy as np
8 import matplotlib.pyplot as plt
9 from img2vec_keras import Img2Vec
10 from sklearn.metrics.pairwise import cosine_similarity
11 from sklearn.decomposition import PCA
12 from sklearn.manifold import TSNE
13 from sklearn.preprocessing import StandardScaler
14 from matplotlib.offsetbox import OffsetImage, AnnotationBbox
15 import cv2
16 import os
17
18 # Загрузка датасета
19 url = 'https://drive.google.com/uc?id=1Madin3qXe3Xexox5q0hASGu1FR2'
20 output = '101_ObjectCategories.tar.gz'
21 gdown.download(url, output, quiet=False)
22
23 # Распаковка архива
24 !tar -xzf 101_ObjectCategories.tar.gz
25 !rm 101_ObjectCategories.tar.gz
26
27 # Инициализация модели для получения векторных представлений
28 img2vec = Img2Vec()
29
30 # Выборка классов для анализа
31 image_classes = ['pizza', 'schooner', 'scissors', 'panda', 'brain']
32 image_paths = []
33
34 # Формирование списка путей к изображениям
35 for image_class in image_classes:
36     image_paths.extend(glob.glob(f'101_ObjectCategories/{image_class}'))
37
38 print(f'Найдено изображений: {len(image_paths)}')
39
40 # Извлечение векторных представлений
41 image_vectors = {}
42 for image_path in image_paths:
43     try:
44         vector = img2vec.get_vec(image_path)
45         image_vectors[image_path] = vector
46     except Exception as e:
47         print(f'Ошибка при обработке {image_path}: {str(e)}')
48
49 # Создание матрицы признаков
50 X = np.stack(list(image_vectors.values()))
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51
52 # Применение PCA для уменьшения размерности
53 pca_50 = PCA(n_components=50)
54 pca_result_50 = pca_50.fit_transform(X)
55 print(f'Объясненная дисперсия для 50 компонент: {np.sum(pca_50.explained_variance_ratio_)}')
56
57 # Дальнейшее уменьшение размерности с помощью t-SNE
58 tsne = TSNE(n_components=2, verbose=1, n_iter=3000)
59 tsne_result = tsne.fit_transform(pca_result_50)
60 tsne_result_scaled = StandardScaler().fit_transform(tsne_result)
61
62 # Подготовка изображений для визуализации
63 images = []
64 for image_path in image_paths:
65     try:
66         image = cv2.imread(image_path)
67         image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
68         image = cv2.resize(image, (50, 50))
69         images.append(image)
70     except Exception as e:
71         print(f'Ошибка загрузки {image_path}: {str(e)}')
72         images.append(np.zeros((50, 50, 3), dtype=np.uint8))
73
74 # Визуализация результатов
75 fig, ax = plt.subplots(figsize=(20, 15))
76 for xy, img in zip(tsne_result_scaled, images):
77     x0, y0 = xy
78     img_box = OffsetImage(img, zoom=0.7)
79     ab = AnnotationBbox(img_box, (x0, y0), frameon=False)
80     ax.add_artist(ab)
81
82 ax.update_datalim(tsne_result_scaled)
83 ax.autoscale()
84 plt.title('Визуализация векторных представлений изображений с помощью t-SNE')
85 plt.show()
86
87 # Сравнение изображений по косинусной близости
88 def compare_images(path1, path2):
89     vec1 = img2vec.get_vec(path1).reshape(1, -1)
90     vec2 = img2vec.get_vec(path2).reshape(1, -1)
91     similarity = cosine_similarity(vec1, vec2)[0][0]
92     return similarity
93
94 # Пример сравнения разных изображений
95 path1 = '101_ObjectCategories/Leopards/image_0001.jpg'
96 path2 = '101_ObjectCategories/Motorbikes/image_0001.jpg'
97 print(f'Разные изображения: {compare_images(path1, path2):.3f}')
98
99 # Пример сравнения похожих изображений
100 path3 = '101_ObjectCategories/Motorbikes/image_0005.jpg'
101 print(f'Похожие изображения: {compare_images(path2, path3):.3f}')

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Collecting git+https://github.com/jaredwinick/img2vec-keras.git
  Cloning https://github.com/jaredwinick/img2vec-keras.git to /tmp/pip-
  Running command git clone --filter=blob:none --quiet https://github.c
  Resolved https://github.com/jaredwinick/img2vec-keras.git to commit 8
  Preparing metadata (setup.py) ... done
Requirement already satisfied: numpy>=1.9.1 in /usr/local/lib/python3.1
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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/
Building wheels for collected packages: img2vec_keras
  Building wheel for img2vec_keras (setup.py) ... done
  Created wheel for img2vec_keras: filename=img2vec_keras-0.2-py3-none-
  Stored in directory: /tmp/pip-ephem-wheel-cache-qnb7i18b/wheels/17/7c
Successfully built img2vec_keras
Installing collected packages: img2vec_keras
Successfully installed img2vec_keras-0.2
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Requirement already satisfied: filelock in /usr/local/lib/python3.12/di
Requirement already satisfied: requests[socks] in /usr/local/lib/pythor
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/site-packages/urllib3-1.26.15-py3-none-any.whl (355 kB)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/site-packages/certifi-2022.9.24-py3-none-any.whl (70 kB)
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in /usr/local/lib/python3.10/site-packages/PySocks-1.7.1-py3-none-any.whl (17 kB)
Downloading...

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To: /content/101_ObjectCategories.tar.gz

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Похожие изображения: 0.900

